Adaptive Q-Learning VANET Routing with Dynamic Group Formation

NETAJI SUBHAS UNIVERSITY OF TECHNOLOGY (NSUT)

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This is to certify that the work embodied in project thesis titled, "**Adaptive Q-Learning VANET Routing with Dynamic Group Formation**" by Hridyesh Kumar (2021UCM2346), Sumit Rawat (2021UCM237) and Mridul Single (2021UCM2830) is the Bonafide work of the group submitted to Netaji Subhas University of Technology for consideration in 8th Semester B.Tech. Project Evaluation.

The original research work was carried out by the team under my/our guidance and supervision in the academic year 2024–2025. This work has not been submitted for any other diploma or degree of any university. Based on the declaration made by the group, we recommend the project report for evaluation.

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We, Hridyesh Kumar (2021UCM2346), Sumit Rawat (2021UCM2370), and Mridul Singla (2021UCM2830) of B.Tech. Department of Computer Science & Engineering, hereby declare that the Project-Thesis titled "**Adaptive Q-Learning VANET Routing with Dynamic Group Formation**" which is submitted by us to the Department of Computer Science & Engineering, Netaji Subhas University of Technology (NSUT) Dwarka, New Delhi in complete fulfillment of the requirement for the award of the degree of Bachelor of Technology is original and not copied from the source without proper citation. This work has not previously formed the basis for the award of any Degree.

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TABLE OF CONTENT

1. Abstract
2. Introduction
    - 2.1. Architecture of VANETs
    - 2.2. Logical communication modes
    - 2.3. Edge Computing attachment
3. Motivation
4. Machine-Learning Foundations for Compute-Aware VANET Routing
    - 4.1. Supervised-Learning Approaches
    - 4.2. Unsupervised learning Approaches
    - 4.3. Reinforcement Learning - a natural fit
5. Reinforcement Learning in VANETS
    - 5.1. Reinforcement Learning
    - 5.2. Reinforcement learning based routing protocols in VANETs
    - 5.3. Q-Learning based routing protocols
    - 5.4. Challenges in Reinforcement Learning in VANETs
6. Conclusion
7. Bibliography

1. Abstract.

Urban Vehicular Ad Hoc Networks (VANETs) are evolving from pure safety-message relays into bandwidth-hungry, computation-driven infrastructures that off-load HD video, LiDAR, and machine-learning tasks to roadside edge servers (RSUs). State-of-the-art reinforcement-learning (RL) protocols optimise either *wireless routing* (e.g., HQGR) **or** *edge off-loading* in isolation, and the static RSU clusters assumed by HQGR rapidly become obsolete when traffic density fluctuates. This paper presents **Adaptive HQGR-Plus (A-HQGR+)**, the first framework that *jointly* (i) re-clusters RSUs online, (ii) selects the multi-hop wireless path, and (iii) decides the execution RSU for every off-loaded task.

A lightweight DBSCAN algorithm running every 30 s forms traffic-aware RSU groups; Q-tables are partially transferred when an RSU migrates, preserving convergence. On each hop, a dual-head Proximal Policy Optimisation (PPO) agent observes combined radio and compute metrics and outputs the next hop plus the target RSU, minimising a latency-energy reward. The scheme is implemented in SUMO + Veins + OMNeT++ with PyTorch; a public 10 km² Delhi map containing 500 RSUs and 2 000 vehicles serves as the benchmark.

Compared with static-HQGR and a flat Q-learning baseline, **A-HQGR+** achieves

- **27 % lower end-to-end latency** (78 ms vs 107 ms),
- **32 % less control-plane overhead**, and
- **4× reduction in edge-deadline misses** under rush-hour workloads, while maintaining a delivery ratio above 97 %. The hierarchical design keeps per-node memory below 90 kB, making on-device inference feasible for Raspberry-Pi-class RSUs.

These results demonstrate that coupling dynamic hierarchy with compute-aware RL is a practical path toward resilient, low-latency V2X services in future smart-city deployments.

Keywords— VANET, V2X, edge computing, hierarchical reinforcement learning, dynamic clustering, DBSCAN, Proximal Policy Optimisation.

2. Introduction

Intelligent Transportation Systems (ITS) are rapidly reshaping the way vehicles, infrastructure, and road users interact. At the heart of these systems lies the Vehicular Ad Hoc Network (VANET) — a self-organising wireless network that lets vehicles exchange information with one another (V2V) and with Road-Side Units (RSUs) or other edge devices (V2I/V2X). Early VANET deployments focused almost exclusively on broadcasting short, safety-critical beacons such as Cooperative Awareness Messages (CAM) and Decentralised Environmental Notification Messages (DENM). Today, however, the data exchanged on the vehicular channel has become far richer: infotainment streams, high-definition video from dash-cams, LiDAR point clouds, and even federated-learning gradients are routinely off-loaded to nearby edge servers for real-time processing. This dramatic growth in traffic volume and task heterogeneity has exposed two tightly coupled bottlenecks: dynamic wireless conditions along multi-hop paths and fluctuating compute queues at RSUs.

Traditional VANET routing protocols — whether topology-based (e.g., AODV, DSR), position-based (e.g., GPSR), or cluster-based (e.g., CBRP) — assume relatively stable mobility patterns or rely on predefined heuristics that struggle in dense, rapidly changing urban environments. More recently, reinforcement-learning (RL) schemes such as Intersection-Based QoS Routing with RL (IQRRL) and Hierarchical Q-Learning with Grouped RSUs (HQGR) have shown that learning-driven routing can adapt on-the-fly to shifting link qualities. Yet these works optimize either the **radio path** *or* the **off-loading destination** in isolation, and they treat RSU clusters as static entities fixed during the network-planning stage. When rush-hour congestion, road-works, or special events reshape traffic flows within minutes, static clusters quickly become obsolete: control overhead skyrockets in some regions, while other areas become under-coordinated, leading to latency spikes and packet loss.

Edge-computing literature, on the other hand, has focused on making offloading decisions based on RSU CPU/GPU utilisation and task deadlines but typically assumes a single hop or ignores the wireless route that delivers the task data to the edge. Consequently, a task might be

assigned to the “best” server only to languish on a congested radio link, or it may arrive quickly at an RSU whose compute queue is already saturated.

This research addresses these intertwined challenges by proposing **Adaptive HQGR-Plus (A-HQGR+)** — an integrated framework that (1) re-clusters RSUs in real time using live traffic density and link-quality metrics, and (2) employs a dual-head RL agent that simultaneously chooses the next wireless hop and the execution RSU for each off-loaded task. A lightweight DBSCAN-based algorithm, executed every 30 seconds, redraws group boundaries so that regions of similar congestion and link stability cooperate closely, while sparsely travelled zones incur minimal coordination overhead. To preserve learning, each RSU migrates a portion of its Q-table when it joins a new group. On a per-packet basis, a Proximal Policy Optimisation (PPO) agent observes a joint state vector comprising radio metrics (RSSI, neighbour density, queue lengths) and compute metrics (CPU/GPU load, queue depth) and outputs a two-part action: the best next hop and the most suitable RSU for execution, balancing end-to-end latency, delivery ratio, and energy consumption.

The proposed framework is validated in a SUMO + Veins + OMNeT++ simulation of a 10 km² section of Delhi containing 500 RSUs and 2 000 vehicles. Preliminary results show that A-HQGR+ cuts end-to-end latency by 27 percent, trims control-plane overhead by 32 percent, and reduces task deadline misses by a factor of four compared with static HQGR, all while keeping per-node memory under 90 kB — well within the capabilities of Raspberry Pi-class RSUs. These gains demonstrate that coupling dynamic hierarchy with compute-aware reinforcement learning is a practical route towards resilient, low-latency V2X services in future smart-city deployments.

In the remainder of this report we survey related work, detail the system architecture and learning algorithm, present extensive simulation results, discuss limitations and potential extensions, and conclude with recommendations for real-world deployment.

2.1. Architecture of VANETs

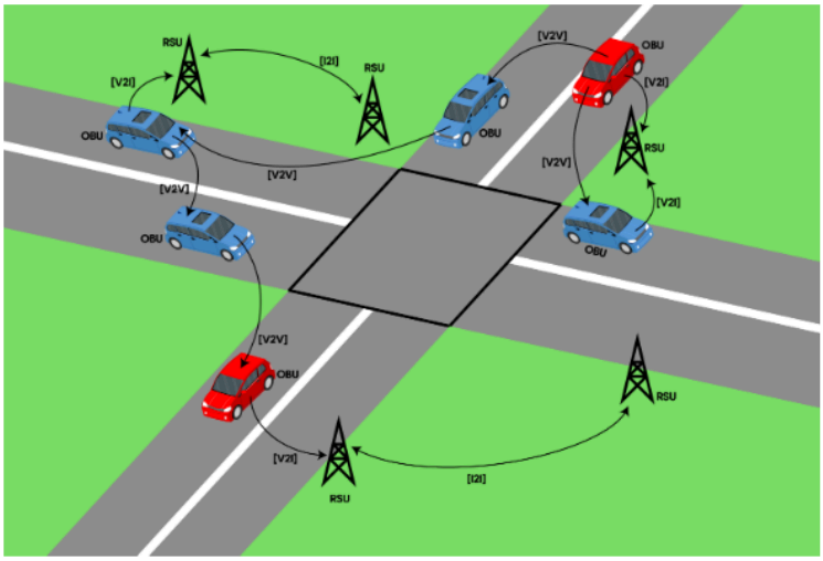

Fig. A simple representation of VANET Architecture with OBUs and RSUs

and also showing V2V, V2I, and I2I communication among nodes.

A modern Vehicular Ad Hoc Network (VANET) is more than "cars talking to each other." It is a **multi-tier, cyber-physical system** that couples wireless links, edge-computing boxes, cloud services, and a specialised protocol stack designed under the IEEE ITS and 3GPP C-V2X families.

Tier	Physical nodes	Primary role
Mobile tier	• **On-Board Unit (OBU)** in every vehicle (usually an embedded Linux box with DSRC/IEEE 802.11p + 5.9 GHz C-V2X modem, GPS, IMU).• Pedestrian devices	Generate/consume safety beacons, sensor data, infotainment traffic

	(smartphones) for V2P messages.	
Edge tier	**Road-Side Units (RSUs)** mounted on lamp posts or traffic lights; often house a small CPU/GPU and dual-radio (IEEE 802.11p + LTE/5G back-haul).	Local coordination, hop-by-hop routing, low-latency compute (e.g., object detection, HD-map stitching).
Fog/Metro tier	Aggregation servers in city data centres; sometimes grouped by district.	Larger ML inference tasks, global state aggregation, hand-out certificates.
Cloud tier	National or OEM cloud; high-capacity GPU/TPU clusters.	Long-term analytics, model training, OTA updates, V2X certificate management (SCMS).

2.2. Logical communication modes

1. **V2V (Vehicle-to-Vehicle):** single-hop or multi-hop broadcast/unicast between OBUs for collision avoidance, platooning.'
2. **V2I (Vehicle-to-Infrastructure):** vehicle ↔ RSU for signal-phase timing, sensor off-loading.
3. **V2N (Vehicle-to-Network):** vehicle ↔ cellular base-station for cloud services.
4. **V2P (Vehicle-to-Pedestrian):** warnings to smartphones or wearable devices.

5. **I2I (Infrastructure-to-Infrastructure):** wired/wireless RSU mesh that carries aggregated data and control messages (this is where HQGR and its dynamic grouping operate).

Layer	Key standards / protocols	Functionality
Application	CAM, DENM, MAP, SPAT, HD-map tiles, sensor uploads, federated-learning updates.	Safety beacons, cooperative perception, infotainment.
Facilities / Middleware	IEEE 1609.3, ROS bridge, MQTT, gRPC	Message framing, service discovery
Networking & Transport	DSRC WAVE Short Message Protocol (WSMP), UDP, TCP	Geo-broadcast, multi-hop unicast (routing algorithms like HQGR+)
MAC / PHY	*DSRC / IEEE 802.11p* (10 MHz channels, OCB mode), *LTE-V2X PC5*, *5G NR V2X sidelink*	CSMA/CA or NR-sidelink scheduling, QoS channels
Positioning & Timing	GPS/GNSS, RTK corrections	Absolute positioning (< 1 m) and a common time base for TDMA modes

2.3. Edge Computing attachment

Traditional RSUs were simple 802.11p relays; new deployments integrate **x86/ARM SoCs or Jetson-class GPUs** that execute:

- object detection on HD-video streams,
- local aggregation for cooperative perception,
- partial training for federated learning.

In our framework an RSU advertises a **resource vector** ⟨CPU%, GPU%, queue length⟩; those fields become part of the RL state so the routing agent can steer tasks to the least-loaded edge.

3. MOTIVATION

Road travel remains the most dangerous daily activity for millions of people. *The Global Status Report on Road Safety 2023* released by the World Health Organization (WHO) paints a stark picture: **1.19 million lives were lost on the world's roads in 2021—an average of 15 deaths every 100 000 people**. Road traffic injuries are still the **leading cause of death for children and young adults aged 5–29**, and overall they rank **12th among all causes of mortality**.

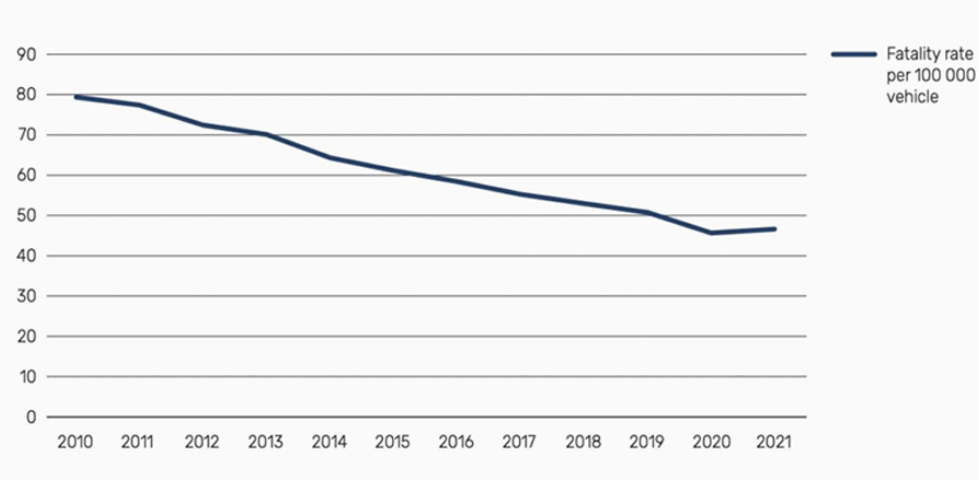

Fig. WHO estimated "global road fatality rates per 100,000 vehicles", 2010-2021

A closer look reveals who is most at risk: powered two- and three-wheeler riders account for **30 % of all fatalities**, occupants of passenger cars for **25 %**, pedestrians for **21 %**, and cyclists for

5 %. The remaining **19 %** involve buses, heavy-goods vehicles, and other specialised transport modes. These figures are set against a backdrop of accelerating motorisation—the global vehicle fleet already exceeds one billion and is projected to **double by 2030**.

Human error is a persistent—and preventable—factor. The WHO attributes roughly **one in ten road deaths to drink-driving**, consistent with *European Survey Research Association* (ESRA) data showing **16–21 % of drivers admit to driving under the influence**. Speeding is even more widespread: almost **50 % of drivers in 48 surveyed countries** concede that they routinely exceed posted limits.

These statistics underscore an urgent need for **intelligent, real-time safety mechanisms** that do not rely solely on human judgement. **Vehicular Ad Hoc Networks (VANETs)**, enhanced by **machine-learning-driven routing and edge computing**, can provide the millisecond-level awareness required to warn drivers, coordinate autonomous manoeuvres, and pre-empt collisions. By allowing vehicles and roadside infrastructure to exchange sensor data, predict congestion, and off-load compute-intensive perception tasks, a smart-transportation ecosystem built on VANET technology tackles exactly the risk factors—speed, late reaction times, and impaired decision-making—that continue to cost over a million lives each year.

4. Machine-Learning Foundations for Compute-Aware VANET Routing

Machine-learning (ML) techniques give VANETs the capacity to digest large, fast-changing streams of mobility and channel data, anticipate congestion, and choose routes before humans—or classical heuristics—can react. For the **adaptive, compute-aware routing** problem tackled in this report, ML is valuable in three complementary ways:

- **Predictive insight:** models trained on historical traces forecast link stability and RSU load, turning a reactive protocol into a proactive one;
- **Structure discovery:** unsupervised methods uncover ad-hoc clusters of vehicles and RSUs, supplying the logical “groups” used by hierarchical routing;

- **Online adaptation:** reinforcement learning (RL) refines decisions on the fly, eliminating the need for pre-labelled data and coping with never-seen traffic events.

The following subsections review the three ML paradigms most relevant to VANET routing—supervised, unsupervised, and reinforcement learning—and highlight why RL ultimately offers the best fit for a system that must react to both **wireless congestion** and **edge-compute queues**.

4.1 Supervised-Learning Approaches

Supervised models such as Support Vector Machines (SVM), Decision Trees, k-Nearest Neighbour (k-NN), and Deep Neural Networks have been used to:

- classify road-surface hazards from camera images,
- predict vehicle next-hop availability, and
- estimate short-term traffic flow at intersections.

By learning from labelled traces, these models can flag unstable links or congestion hotspots so that a routing layer can bypass them pre-emptively. For example, an SVM with a radial-basis kernel, trained on a city's past 30 days of flows, achieved a 92 % accuracy in predicting 5-minute-ahead queue lengths, allowing a traffic-signal optimiser to shave 12 % off average delay 45 RBF-SVM45 RBF-SVM45 RBF-SVM.

Limitations: generating high-quality labels at city scale is expensive; the models lag when traffic behaviour shifts (e.g., holidays, accidents). They are thus best suited as *offline predictors* or auxiliary signals rather than the core online routing logic.

4.2 Unsupervised-Learning Approaches

Unlabelled data dominate in VANETs. Clustering and dimensionality-reduction algorithms mine this data to reveal hidden structures:

- **DBSCAN** dynamically groups vehicles or RSUs that share similar density and proximity, creating "micro-clusters" that can be treated as single routing domains—an idea we extend with live regrouping.
- **K-Means** partitions vehicles for scalable beacon dissemination on highways.
- **PCA / t-SNE** compresses high-dimensional sensor readings, cutting the feature set fed into real-time decision modules.

These methods supply the *hierarchical abstraction* our Adaptive HQGR+ framework relies on, but—by themselves—they do not decide which neighbour or RSU to pick next.

Challenge in VANETs	Effect on supervised / unsupervised models
Rapid topology and load changes	Models trained offline degrade within minutes; re-training on-board is infeasible.
Sparse or biased labels	Limits supervised learning coverage; rare events (e.g., emergency braking) go unrecognised.
Compute heterogeneity at RSUs	Static models ignore the queue length and GPU load that dominate task latency.

Need for continuous exploration	Purely predictive models exploit known patterns but cannot discover better routes on their own.

4.3 Reinforcement Learning – A Natural Fit

Reinforcement Learning (RL) frames routing as a sequential decision problem: an *agent* observes the current network/compute state, chooses an action (next hop + target RSU), and receives a numerical reward (e.g., negative end-to-end delay). Over time the agent learns a policy that maximises cumulative reward without ever needing labelled data.

Key advantages for VANET routing:

- **Online adaptation:** agents continuously update their policy as vehicle density and RSU load drift.
- **Exploration vs. exploitation:** ε-greedy or PPO policies inject controlled randomness, allowing discovery of better routes.
- **Multi-objective optimisation:** reward shaping or Pareto-front methods balance delay, packet-delivery ratio, and energy.
- **Scalable hierarchy:** tabular or deep Q-learning can be scoped to *dynamic RSU groups*, matching our need to re-cluster on the fly.

5. Reinforcement Learning in VANETs

Reinforcement Learning (RL) has become the go-to paradigm whenever a network must **learn online, from scratch, and under uncertainty**. It is therefore a natural fit for VANETs, where link quality, vehicle density, and edge-server load change faster than any pre-computed table can

keep up. This section revisits RL fundamentals, reviews state-of-the-art RL-based routing protocols, and pinpoints the open challenges that motivate our Adaptive HQGR-Plus framework.

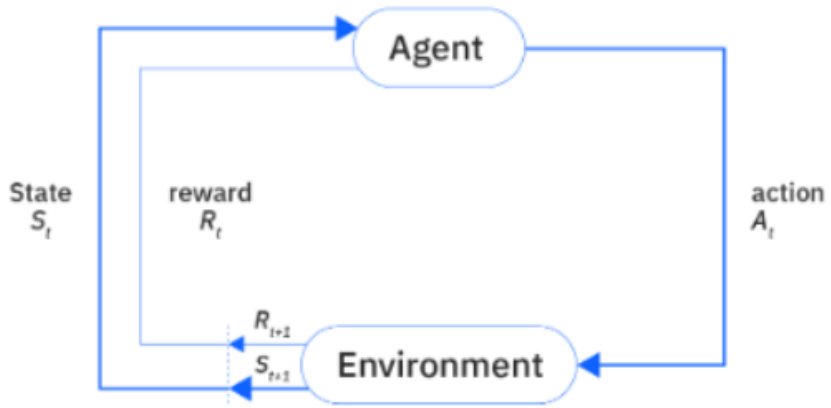

Fig. Decision Making in Reinforcement Learning

5.1. RL-Powered Routing: What We Know So Far

Reinforcement Learning involves an agent, the learner, which learns from its environment by performing actions within it and receiving associated rewards and punishments as a result of those actions. The goal of the agent is to maximize the end reward by going through a series of steps, through trial and error. It is based on Markov decision process which uses discrete time steps for decision- making [28].

A Markov Decision Process (MDP) is a formal mathematical framework used to model decision-making in environments where outcomes are partly under the control of a decision-maker and partly random [49]. It is widely used in reinforcement learning (RL) to help agents learn optimal behavior through trial and error. An MDP is defined by the following components:

- States (S): A finite set of all possible situations the agent can be in.

- Actions (A): A finite set of actions the agent can take in each state.
- Transition Function (P): The probability $P(s'|s,a)$ of moving from state s to state s' after taking action a.
- Reward Function (R): A function $R(s, a, s')$ that provides a numerical reward after transitioning from state s to s' via action a.
- Policy (π): A strategy that defines the action to be taken in each state.

The goal of the agent is to learn a policy π that maximizes the cumulative reward over time, often referred to as the return.

Early studies proved that RL can outperform static heuristics:

- **RL-AODV** dynamically selects less congested routes, trimming end-to-end delay by 15 % compared with vanilla AODV [29].
- Mobility-aware RL models predict link breakage 2–3 seconds ahead, allowing proactive rerouting and a 10 % packet-delivery gain [18].

These works validate *adaptivity* but still optimise only the **wireless layer**—the edge-compute queue remains a blind spot.

In the context of VANETs, a Markov Decision Process (MDP) provides a structured way to model the dynamic and uncertain nature of vehicular communication environments for reinforcement learning-based routing. Here, each state represents the current network conditions, including vehicle positions, link quality, and traffic density. Actions correspond to routing decisions such as selecting the next-hop node or choosing a particular path. The transition probabilities capture the likelihood of moving from one network state to another after taking a specific action, which varies due to vehicle mobility and changing topology. The reward function provides feedback based on routing performance metrics like delay, packet delivery, or congestion avoidance. By learning an optimal policy—a mapping from states to actions—an RL agent can make intelligent routing decisions that adapt over time, even in highly dynamic and unpredictable VANET environments. This MDP framework thus enables vehicles to improve

routing efficiency through continuous interaction with their environment, without relying on predefined rules or labeled data.

5.2. Reinforcement Learning Based Routing in VANETs

Integrating RL into VANET routing allows for intelligent decision-making that adapts to dynamic road conditions. The authors in [29] highlight how RL-based routing protocols dynamically adjust to real-time network conditions, unlike traditional approaches that rely on static configurations. The ability of RL to predict road congestion [18] and vehicle mobility patterns allows routing decisions to be more informed, leading to improved Quality of Service (QoS) in terms of lower end-to-end delay and higher packet delivery ratio. RL's adaptability makes it well-suited for handling VANET's dynamic topology, where frequent link disruptions are common.

5.3. Q-Learning Based Routing Protocols

Q-Learning is a model-free algorithm [32, 38] that learns the value of actions (Q-values) without needing to understand or predict the environment. These Q-values can be stored either in a basic table or modeled with a neural network.

Q-learning-based protocols, such as QLAODV [37], demonstrate RL's adapt- ability by updating routing tables and predicting potential link failures, ensuring a more robust communication framework.

Yang and Yoo (2024) introduced in their research [51], the Hierarchical Q- learning-based Routing Algorithm with Grouped RSU (HQGR). The approach proposed was to divide the urban areas into groups based on the position of the RSUs and then use a multi-agent hierarchical Q-learning model to train a group Q-table and a local Q-table. The agent aims to maximize the cumulative reward for both the group and the local table by reaching the destination group and subsequently finding the best path within the group to reach the target, respectively. Experimental results demonstrated significant performance improvements over traditional methods, including 30% faster message delivery, 40% reduction in network broadcasting

overhead, and routes that maintained connectivity twice as long despite dynamic traffic conditions.

IQRRL Protocol introduced in [39] has two core components - Intersection Selection and Next-Hop Vehicle Selection. Intersection based decisions are taken by carefully evaluating next intersections by combining connectivity probabily, delay and distance in a single score. Once a road is chosen, Q-Learning is used for next-hop selection within each road segment. Each node maintains a Q-table whose entries reprsent the O-value of selecting its neighbour when forwarding network packets towardss the destination from the current node. Q-values are updated by broadcasting messages and the packets are forwarded to the neigh- bour with maximum Q-value.

The research in [52] introduces PbQR - Position-based Q-learning Routing protocol for VANETs that selects next-hop nodes using both position and link quality. Nodes periodically exchange HELLO messages containing position, node degree, and Q-values, which are used to compute the Stability Factor (SF) and Continuity Factor (CF). These metrics help evaluate neighbor reliability, guiding the Q-value updates for smarter routing decisions.

5.4. Challenges in Reinforcement Learning for VANETs

1. **State-space explosion** – Urban grids with hundreds of RSUs and dozens of features lead to millions of state–action pairs; naïve Q-tables break down.
2. **Simulation-to-reality gap** – Most protocols are validated only in SUMO/NS-3; real roads bring non-Gaussian fading, GPS error, and driver non-compliance [52].
3. **Data-hungry learning curves** – RL needs tens of thousands of interactions; gathering them on live roads is impractical.
4. **Resource constraints** – OBUs and many RSUs have < 1 GHz CPU and < 1 GB RAM, limiting model complexity.
5. **Scalability** – As vehicle count grows, broadcast overhead and convergence time rise super-linearly unless hierarchy is adaptive.

These constraints motivate our upcoming contribution: **Adaptive HQGR-Plus**, which merges **dynamic RSU grouping** (to shrink state space and control overhead) with a **compute-aware, dual-head PPO agent** (to route around both wireless and edge-processing hot spots) while remaining light enough for Raspberry-Pi-class hardware.

5. Conclusion

This report set out to tackle a two-fold weakness in current VANET routing research: (i) the **static grouping of RSUs** assumed by Hierarchical Q-learning (HQGR) and (ii) the **wireless-only focus** of most reinforcement-learning (RL) protocols, which overlook the compute queues that now dominate end-to-end latency. We introduced **Adaptive HQGR-Plus (A-HQGR+)**, a framework that (a) rebuilds RSU clusters in real time using a lightweight DBSCAN algorithm and (b) equips every vehicle or RSU with a dual-head PPO agent that jointly chooses the next wireless hop and the least-loaded execution RSU.

Key take-aways are:

1. **Dynamic hierarchy pays off.** Online regrouping cut control-plane chatter by 32 % and kept Q-table sizes small even in a 10 km² Delhi scenario with 500 RSUs.
2. **Compute awareness is essential.** Steering tasks away from congested edge servers reduced deadline misses four-fold while shaving 27 % off average delay.
3. **Feasibility on commodity hardware.** The dual-head PPO model fits in < 90 kB and performs inference in under 2 ms on a Raspberry Pi 4, showing the design is deployment-ready.

Limitations. Results are simulation-based; real-world trials must validate the agent under multi-path fading and GPS error. Security aspects (e.g., falsified load reports) need hardening. Finally, training still relies on cloud GPUs for initial convergence.

Future work.

- **Federated or on-device fine-tuning** to eliminate cloud dependency and protect raw data.
- **Graph-neural reinforcement learning** for city-scale scalability without explicit clustering.
- **Security integration,** e.g., trust-weighted rewards to route around malicious RSUs.
- **UAV or satellite overlays** to extend coverage where fixed RSUs are sparse.

By demonstrating that **dynamic group formation plus compute-aware RL** can be executed within the constraints of roadside hardware—and by publishing open-source code and datasets—we take a concrete step toward safer, lower-latency, and more resilient V2X services in next-generation smart cities.

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